Countering the Winner's Curse in Aggregated-Signal, Common-Value Auctions in the Sequential and Non-Sequential Settings

Jessica Chen[1]
Harvard University
jessica_chen@college.harvard.edu

Dean Huang[1]
Harvard University
deanhuang@college.harvard.edu

***Abstract*—Common-value auctions are known for the winner's curse phenomenon, where the highest bidder often ends up overpaying due to overestimation of the item's true common value. While literature has canonically analyzed auction designs and equilibria in the maximum signal setting proposed by Brulow and Klemperer (2002), we are interested in providing exploratory analysis in the aggregated signal setting, where the common value is the sum of the bidders' signals.**

This paper provides an analysis of this aggregated signal common-value auctions under both symmetric (non-sequential) and asymmetric (sequential) settings. In the symmetric setting, our main contributions include quantifying the impact of the winner's curse, finding pure Nash equilibria in toy games, and proposing a novel inclusive posted price auction design to eliminate the winner's curse.

In the asymmetric setting, we show that, in equilibrium, no bidder ever bids more than twice their own private signal, even though naive benchmarks would suggest significantly higher bids. We further investigate how these findings scale as $k \to \infty$, how continuous approximations might inform the discrete models, and how additional information or mechanism design tweaks could influence outcomes.

I. Introduction

A common-value auction sells an item that is valued the same across all bidders, but the true common value is unknown. Each bidder receives a private signal about this value. Classic examples include auctions for oil drilling rights, financial assets, and spectrum licenses. The well-known *winner's curse* arises when the highest bidder's estimate is overly optimistic; upon winning, it becomes evident that other bidders' signals were likely lower, leading to a painful overpayment.

Prior literature frequently focuses on the maximum-signal model, in which the common value equals the highest of the bidders' signals.

$$v(s_1, \ldots, s_N) = maxs$$

This model exacerbates the winner's curse by providing a sharp upper bound on the common value, making overestimation more severe.

Our novel contribution is rooted in our interest in the aggregated-signal model, where the common value is the sum of all bidders' signals.

$$v(s_1, \ldots, s_N) = \sum_{i=1}^{N} s_i$$

The aggregated signal model is relevant in applications with crowdsourced innovation, such as the case where a company auctions off rights to develop a product based on aggregate of intellectual contributions (e.g. data). Each bidder has private data, which informs the potential success of innovation. This model can also be easily extended to a mean signal model.

In this paper, we present a unified analysis of common-value auctions in both symmetric (non-sequential) and asymmetric (sequential) settings:

- **Symmetric Setting (Non-Sequential):** Multiple bidders simultaneously submit bids. We consider scenarios with signals drawn from distributions over $[\underline{s}, \bar{s}]$ and also discrete models (e.g., rolling a k-sided die). We analyze standard auctions (e.g., first-price, second-price) and discuss the pure Nash equilibria. We also propose a novel inclusive posted price mechanism to eliminate the winner's curse.
- **Asymmetric Setting (Sequential):** Two bidders each receive an independent k-sided private signal, and the common value is their sum. The auction proceeds sequentially: one bidder places a bid first, and the other responds. This asymmetry provides additional inference opportunities and alters equilibrium conditions, shedding light on how the timing and observation of bids influences strategies.

II. Related Literature

The winner's curse has long been a central theme in the economics literature on auctions with common values. Classic analyses, such as those by Milgrom and Weber, show how information structures influence equilibrium bidding. The maximum-signal model is known to exacerbate the winner's curse, and much of the literature uses it as a benchmark. In contrast, the aggregated-signal model, though less extreme, still poses significant inference challenges.

Bulow and Klemperer (2002) demonstrate scenarios where a posted-price mechanism can outperform standard auctions, highlighting the importance of understanding equilibrium bid adjustments under uncertainty. Bergemann, Brooks, and Morris (2019) show that standard auction formats may yield similar outcomes in certain common-value settings.

III. Symmetric Environment

Consider an auction with N bidders, indexed by $i \in \{1, \dots, N\}$. Each bidder i receives a signal $s_i \in [\underline{s}, \bar{s}]$ about the good's common value. These signals are independent draws from an absolutely continuous cumulative distribution F with strictly positive density f. The common value of the good is:

$$v(s_1, \dots, s_N) = \sum_{i=1}^{N} s_i.$$

The item's value is common but unknown, and each bidder receives private signals drawn from some underlying distribution.

All bidders submit their bids b_i simultaneously. The highest bid wins, and the winner's utility is $v - b_{\text{winner}}$. Other bidders get zero utility. In the case of a tie, the winner is chosen at random with equal probability. We assume all signals and bids are whole numbers.

In standard formats such as first-price or second-price auctions, equilibrium analysis must account for the winner's curse, since winning reveals that other signals are likely lower than one's unconditional expectation. As our default setting, we are interested in the first-price auction.

We also consider a discrete version as a toy model: suppose $N = 2$ and each bidder's signal is drawn by rolling a k-sided die with faces $\{1, 2, \dots, k\}$. This simplification helps us explore the existence of pure Nash equilibria, understand how the winner's curse magnitude δ depends on k, and investigate mechanism design variations such as the inclusive posted price mechanism.

IV. First-Price Auction and the Winner's Curse (Symmetric)

We work through toy games to better understand how this modification would change the results of the Bergemann 2020 paper.

A. *Magnitude δ of Winner's Curse*

The winner's curse occurs in non-equilibrium settings, when bidders bid based on their interim expectation instead of their posterior expectation.

As a reminder, the interim expectation is the expectation of common value, conditioned only on Bidder 1's private signal s_1.

$$\mathbb{E}[v|s_1] = s_1 + \mathbb{E}[s_2] = s_1 + \frac{1}{k}\sum_{i=1}^{k} i$$

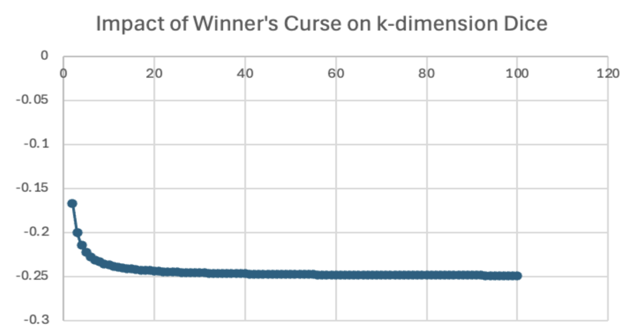
The posterior expectation is the expectation of the common value, conditioned on s_1 being the winning bid.

$$\begin{aligned}
&\mathbb{E}[v|s_1, \text{win}] \\
&= \frac{\mathbb{P}[v \cap \text{win}] \cdot v}{\mathbb{P}[\text{win}]} \\
&= \frac{\mathbb{P}[s_1 > s_2] \cdot \mathbb{E}[v|s_1 > s_2] + \frac{1}{2}\mathbb{P}[s_1 = s_2] \cdot \mathbb{E}[v|s_1 = s_2]}{\mathbb{P}[s_1 > s_2] + \frac{1}{2}\mathbb{P}[s_1 = s_2]}
\end{aligned}$$

The impact of the winner's curse, δ, exists because new information about the common value is revealed upon winning.

B. *Toy Games in the Symmetric Setting*

The toy game setup assumes a 2nd-price auction with two players, each rolling a die to get their independent private signal. We consider this scenario for dice with k sides. Bidder i receives signal s_i and bids b_i. In the event of a tie, payoff of all bidders is 0.



It's clear that δ of $s_1 = k$ converges asymptotically to -0.248 as k increases. View the GitHub repo of the simulation here.

We unfortunately were not able to analytically find the reason for this asymptotic relationship. We will leave this for next steps.

V. Pure Nash Equilibria in the Symmetric Setting

A. *Pure Nash Equilibrium for $k = 2$ Die*

Let b_j^* refer to the pure equilibrium bid given a signal $j \in k$.

- $(b_1^*, b_2^*) = (1, 2)$: If $(s_1, s_2) = (1, 2)$, then bids are $(1, 2)$, payoffs $(0, 1)$. b_1 can deviate to 2 and receive potential payoff ≥ 0.
- $(b_1^*, b_2^*) = (2, 3)$: If $(s_1, s_2) = (1, 2)$, then bids are $(2, 3)$, payoffs $(0, 0)$. b_2 can deviate to 2 and receive potential payoff ≥ 0.
- $(b_1^*, b_2^*) = (2, 2)$: Unique PNE. Even if both players were to roll 2, a deviation to bid higher to 3 would result in the same expected payoff of 1. Bidding even higher would result weakly worse payoffs.

Since the virtual values are monotonically increasing, we know that the highest-signal bidder will receive allocation. As such, we can see that $(2, 2)$ is the unique PNE.

B. Pure Nash Equilibrium for $k = 3$ Die

We claim that there is no PNE for dice with $k = 3$ sides.

Theorem 1. *Each unique signal s_j has a unique equilibrium bid b_j^*, regardless of the strategy space k.*

We can show this inductively. In the base case, we have a $k = 2$-sided die, which we have shown above to have a unique PNE. Say that we have solved for the PNE bids for a k-sided die.

Then let us consider the PNE for a $k+1$-sided die. Myerson (1981) has shown that, for settings with independent private values and when utility functions are separable in their signals, when the bidder's virtual value is weakly increasing in their signal, then the optimal allocation is also interim monotonic. We also understand this through the direct mechanism. As such, for a bidder i with a signal $s_{i,j} \in k$, bidder i knows they can only win over other bidders for which $s_{i-1,j} \leq s_{i,j}$. Bidder i is not concerned for the set of bidders with signals $> s_{i,j}$, because they would lose; losing results in utility zero, to which they would be indifferent. Therefore, it does not matter what the upper bound of the opposing bidder's signal is, and thus it does not matter what k is; their bid is only dependent on possible $b_j^* \leq s_j$.

As such, given we have access to the PNE bids for a k-sided die, then in finding the equilibrium for a $k+1$-sided die, we only need to find b_{k+1} given $b_1^*, \ldots, b_j^*$ for $j \in k$. Therefore, we can find the PNE using casework (for $b_3^* \in [2, 5]$ since bids must be monotonically increasing in signal and $b_3^* = 6$ would yield 0 payoff).

- $(b_1^*, b_2^*, b_3^*) = (2, 2, 2)$: If $(s_1, s_2) = (3, 3)$, then either player can gain utility by deviating to bid 4.
- $(b_1^*, b_2^*, b_3^*) = (2, 2, 3)$: If $(s_1, s_2) = (3, 3)$, then either player can gain utility by deviating to bid 4.
- $(b_1^*, b_2^*, b_3^*) = (2, 2, 4)$: If $(s_1, s_2) = (2, 3)$, then first bidder can deviate to bid 4, which increases their expected payoff from 0 to 0.5.
- $(b_1^*, b_2^*, b_3^*) = (2, 2, 5)$: If $(s_1, s_2) = (2, 3)$, then second bidder can deviate to bid 4, which increases their expected payoff from 0 to 1.

Therefore, we have shown that no PNE exists for 3-sided die.

C. Pure Nash Equilibrium for $k > 3$ Die

Theorem 2. *Dice with sides $k \geq 3$ do not have a pure Nash equilibrium.*

Given Theorem 1, we know that if $k = 3$ does not have a PNE, and PNEs are inductible subproblems by their signal, then we also see that $k > 3$ does not have PNEs.

VI. Direct Mechanism

Now we want to talk about mechanism design to increase revenue or counter the winner's curse. By Myerson's revelation principle (1981), we can scope the rest of our discussion to direct mechanisms without loss of generality.

Bidder i's utility from reporting signal $s_i^{'}$ given true signal s_i is

$$u_i(s_i, s_i') = \int_{s_{-1} \in S} q_i(s_i', s_{-1}) v_i(s_i, s_{-1}) f_{-i}(s_{-1}) ds_{-i} - t_i(s_i')$$

The direct mechanism is incentive compatible given $u_i(s_i) = \max_{s_i'} u_i(s_i, s_i')$ for all i and $s_i \in S$; in other words, a bidder bids their true signal. The mechanism is individually rational if $u_i(s_i) \geq 0$.

VII. Inclusive Posted Prices (Symmetric)

Bergemann, Brooks, and Morris (2002) proposed the inclusive posted price auction as a way to counter the winner's curse and boost revenue for the maximum signal model. We will reiterate their argument, then show why this does not work for the aggregated signal model. Finally, we briefly discuss how to potentially modify this auction to work with the aggregated signal model.

A. Revenue Ranking Approach to Show Revenue Boost and Eliminated Winner's Curse for the Maximum Signal Model

[Inclusive posted price mechanism] Consider bidder i_L with the lowest possible signal. The *inclusive posted price* is this bidder i_L's interim expectation of the common value, which is also the largest price such that all are willing to buy the object. In other words, the inclusive posted price is the interim expectation that a bidder i with the lowest possible signal, or p_I such that

$$p_I = \int_{x=\underline{s}}^{\bar{s}} x d(F^{N-1}(x))$$

Theorem 3. *The inclusive posted price yields a higher revenue than the monotonic pure-strategy equilibrium of any standard auction.*

Bergemann et al (2002) prove this by building off of conclusions from two previous work:

1) Bulow and Klemperer (2002) established that a specific posted price mechanism can achieve higher revenue than the English auction with monotonic equilibria.
2) Bergemann et al. (2019) showed that all standard auctions share revenue equivalence for the maximum signal model.

We will briefly summarize the proof for second-price auctions to illustrate later points. In Bulow and Klemperer's posted price mechanism, the object is allocated with uniform probability among bidders who declare their willingness to pay the inclusive posted price p_I for the object, given the highest $N-1$ independent draws from the signal distribution F.

For N bidders, the second-price auction always excludes the highest signal (since the second-highest signal is used for payment). The inclusive posted price revenue is generated by randomly excluding one draw from N and then taking the highest of the remaining $N-1$ draws. This ensures that the inclusive posted price revenue is always higher than the second-price auction revenue, as the highest of $N-1$ is greater than or equal to the second-highest of N.

Furthermore, the inclusive posted price auction's allocation rule assigns an object to a bidder i with equal probability $q_i(s) = \frac{1}{N}$. Because this random assignment does not provide any additional information on the common value between the interim and posterior phases, the expectations for the common value between these two phases do not change based on conditionality. As such, there is no winner's curse.

B. Revenue Ranking Approach in the Aggregated Signal Model

Unfortunately, we have not been able to find an analogous paper to Bergemann et al. (2019) to show revenue equivalence between all standard auctions for the aggregated signal model (or derive it ourselves). As such, we will investigate how the inclusive posted price mechanism impacts results for the aggregated signal model in the second-price auction setting, then in the first-price auction.

1) **Second-price auction.** We propose the same inclusive posted price auction to boost revenue and eliminate the winner's curse compared to the second-price auction. By the revelation principle, we will work under the assumption that all bidders bid their signals. As such, we can keep p_I as the expected value of the maximum of the signals across $N-1$ bidders, excluding one bidder at random. While the common value is different, this does not impact the dynamics of who the winner is and how much they would pay. The random allocation of the object to a participating bidder still does not reveal additional information in the posterior period. As such, by the same logic, the revenue for this inclusive posted price auction is greater than that of the second-price auction, and the winner's curse is eliminated.
2) **First-price auction.** We propose a slightly altered version of the inclusive posted price auction. Similarly, we leverage the revelation principle to consider just the direct mechanism. In the standard first-price auction, the collected revenue is equal to the maximum of the signals across all N bidders. In the revised posted price auction, the posted price is instead the expected value of the maximum of all N bidders, or p_I such that

$$p_I = \int_{x=\underline{s}}^{\bar{s}} x d(F^N(x))$$

The rest of the mechanism is consistent: bidders may each individually choose whether to participate based on the posted price, and the object is allocated at random to the participating bidders. While we obviously have not reached a boosted revenue (in both auctions, the revenue is simply the expected value of the maximum signal across all bidders), we are able to eliminate the winner's curse by removing information gains in the posterior expectation.

VIII. Asymmetric Environment

A. Environment and Signals

We have two bidders, indexed by $i \in \{1, 2\}$. Each bidder observes a private signal:

$$s_i \sim \text{Uniform}\{1, 2, \ldots, k\}, \quad i \in \{1, 2\}.$$

The common value:

$$V = s_1 + s_2,$$

is the sum of the signals. The unconditional expected value of V, before seeing any signal, is:

$$\mathbb{E}[V] = \mathbb{E}[s_1] + \mathbb{E}[s_2] = (k + 1).$$

From bidder i's perspective, knowing s_i, the conditional expectation is:

$$\mathbb{E}[V \mid s_i] = s_i + \frac{k+1}{2}.$$

B. Auction Format

We focus on a sequential ascending auction format:

1) Bidder 1 observes s_1 and chooses a bid b_1.
2) Bidder 2 observes b_1 and knows s_2. Bidder 2 decides whether to outbid by bidding $b_2 > b_1$ or drop out.
3) The highest bidder pays their bid and wins the item, with utility $V - b_{\text{winner}}$.

The key strategic consideration is that each bidder does not know the other's signal but updates beliefs based on the observed behavior (especially in the sequential case).

C. Simultaneous vs. Sequential Settings

In a non-sequential (second-price or first-price) simultaneous auction, both bidders submit bids simultaneously. Equilibrium analysis reveals that both bidders must underbid their naive expectations to avoid negative payoffs.

In the sequential setting, the presence of an "anchor" bid by bidder 1 gives bidder 2 extra information. Bidder 2 uses b_1 to infer something about s_1, and thus about V. Anticipating this, bidder 1 must choose b_1 to avoid being exploited by bidder 2.

D. Winner's Curse Intuition

If a bidder bids precisely their unconditional expectation $\mathbb{E}[V|s_i]$, then winning implies that their bid was higher than any bid the opponent was willing to place, strongly suggesting that the opponent's signal is lower than average. This conditional on-winning expectation is therefore strictly lower than the unconditional expectation. The difference between these two expectations is the *winner's curse* adjustment, which we denote by $\Delta(s_i, k)$:

$$b_i^*(s_i) = s_i + \frac{k+1}{2} - \Delta(s_i, k).$$

Identifying or bounding $\Delta(s_i, k)$ is crucial to understanding equilibrium strategies.

—

IX. Asymmetric Equilibrium Characterization for Small k

Consider $k = 2$. Each bidder's signal is either 1 or 2, each with probability 1/2. There are four equally likely states: $(s_1, s_2) \in \{(1,1),(1,2),(2,1),(2,2)\}$. The common values in these states are $2, 3, 3, 4$, respectively. From bidder 1's viewpoint, if $s_1 = 2$, the unconditional expectation is 3.5. If bidder 1 naively bids 3.5, winning suggests bidder 2 did not top that bid, likely meaning s_2 is closer to 1 than to 2. Empirically, one finds a stable equilibrium where bidder 1 bids around 3 when $s_1 = 2$. For $s_1 = 1$, bidding at or near 2 can make sense. The net effect is a uniform downward shift by about 0.5 for the high signal case.

For $k = 3$, signals are uniformly distributed in $\{1, 2, 3\}$. The unconditional expectation given $s_1 = 3$ is $3 + 2 = 5$, while a naive bidder might consider bidding near 5. However, simulations indicate that equilibrium bidding is lower. Suppose that if $s_1 = 3$, the bidder might bid around 4.5. When $s_1 = 2$, the unconditional expectation is $2+2 = 4$, but equilibrium bids are around 3 or 3.5. For $s_1 = 1$, equilibrium bids become even smaller, potentially around 2.5 or less. These patterns hint at a general phenomenon: as s_1 increases, the gap between the naive expectation and the actual equilibrium bid narrows, but never vanishes. Lower signals face an even more pronounced winner's curse penalty.

Extensive numerical experimentation confirms that for small k, equilibrium strategies can be approximated by:

$$b_1^*(s_1) \approx s_1 + \frac{k+1}{2} - c$$

for some constant c that depends on k. The constant c is not uniform across all signals but tends to be on the order of 0.5 to 1.5 for these small cases. This adjustment factor c reflects the expected "overestimation" that a naive bidder would suffer from if they bid their unconditional expectation.

A. Bidder 2's Best Response in Detail

Given a strategy $b_1^*(\cdot)$ for bidder 1, consider bidder 2's decision after seeing b_1. Bidder 2 observes b_1 and has signal s_2. Let $G(b_1) = \mathbb{E}[s_1|b_1]$ represent bidder 2's posterior on s_1. To outbid b_1, bidder 2 must believe:

$$s_2 + G(b_1) - b_1 \geq 0.$$

If b_1 is known to be monotone in s_1, then $G(b_1)$ is just the inverse mapping $(b_1^*)^{-1}(b_1)$. Assume monotonicity and invertibility. If $b_1^*(s_1)$ is strictly increasing, then:

$$s_1 = (b_1^*)^{-1}(b_1),$$

and hence

$$G(b_1) = (b_1^*)^{-1}(b_1).$$

To solve for equilibrium, one can imagine guessing a form for $b_1^*(s_1)$, plug it into bidder 2's inequality, and solve for conditions that leave bidder 1 indifferent. In equilibrium, bidder 1 sets b_1 so that their expected payoff is zero at the margin. This means:

$$\mathbb{E}[(s_1 + s_2 - b_1)\mathbf{1}(\text{not outbid})|s_1] = 0,$$

which implies:

$$b_1 = \mathbb{E}[s_1 + s_2|s_1, \text{win event}] = s_1 + \mathbb{E}[s_2|\text{win event}].$$

The event of winning is correlated with s_2 being sufficiently small, since if s_2 were large, bidder 2 would outbid. Thus the conditional expectation $\mathbb{E}[s_2|\text{win event}] < \frac{k+1}{2}$, providing the negative correction term (the winner's curse).

X. Inductive Proof for the Asymmetric Bid Bound

Theorem 4. *In any equilibrium of the described game (asymmetric or not), no bidder i ever bids more than $2s_i$.*

Proof. Base Case: $k = 1$. With only one side on the die, $s_i = 1$ always, and the common value is $V = 2$. The only equilibrium is both bidders bidding 2, which equals $2s_i$. Thus, for $k = 1$, the statement holds trivially.

Inductive Step: Assume that for any $(k-1)$-sided version of the game, no equilibrium bid exceeds $2s_i$. Now consider the k-sided game. Suppose for contradiction there is an equilibrium with a bidder i and a signal s_i such that $b_i > 2s_i$. Write $b_i = 2s_i + \delta$ for some $\delta > 0$.

If a bidder attempts to bid this high, consider the following: 1. The other bidder, upon seeing such an extreme strategy, can respond either by outbidding when profitable or by letting the first bidder win at a high price. If the second bidder's signal is at least medium-sized, they can exploit the over-bidder. If the second bidder's signal is small, letting the over-bidder win at a price $> 2s_i$ ensures the over-bidder's expected payoff is negative.

2. By the inductive hypothesis, in a $(k-1)$-sided game, such behavior (bidding above $2s_i$) is not equilibrium. Adding one more side increases the unconditional expectation slightly, but it does not remove the winner's curse. If anything, higher k makes the distribution of s_2 more spread out, enabling even more selective responses by the opponent.

3. Since equilibrium requires no profitable unilateral deviations, the existence of such a high bid strategy would prompt profitable deviations by the opponent. Hence it cannot be stable.

By contradiction, no equilibrium has $b_i > 2s_i$. Thus, by induction, the claim holds for all k. □

XI. Implications of the Asymmetric Bid Bound

This result is striking because it sets a universal upper bound on equilibrium bids that scales linearly with one's own signal. Intuitively, even if you know that on average $V = s_i + \frac{k+1}{2}$, you dare not bid close to this value in equilibrium. The winner's curse is so strong that you must remain well below the naive benchmark, confined within a linear factor of s_i.

Given that s_i itself grows linearly with k, one might wonder how large k affects the gap between $2s_i$ and $s_i + \frac{k+1}{2}$. For large k, $\frac{k+1}{2}$ can dwarf $2s_i$ for small s_i. For example, if $s_i = 1$ and $k = 100$, the unconditional expectation is $1 + 50.5 = 51.5$, while the upper bound says no bidder can bid above 2. This extreme discrepancy underscores how conservative equilibrium behavior becomes for small signals, especially as k grows large. The larger the range of s_2, the more severe the winner's curse for lower signals.

On the other hand, if s_i is large (close to k), say $s_i = k$, then $2s_i = 2k$. The naive expectation would be $k + \frac{k+1}{2} = k + \frac{k}{2} + \frac{1}{2} \approx 1.5k$, which is actually less than $2k$. For very large signals, the upper bound $2s_i$ is not the binding constraint, and the equilibrium bid might be closer to $s_i + \frac{k+1}{2} - \Delta$, where Δ might be moderate. This explains why high-signal types are less drastically constrained than low-signal types by the winner's curse.

XII. Continuous Approximations for Asymmetric Setting

As $k \to \infty$, we can approximate the discrete uniform distribution by a continuous distribution on $[0, 1]$ or $[1, k]$. For simplicity, consider $s_i \in [0, 1]$ uniformly. Then:

$$\mathbb{E}[V|s_1 = x] = x + 0.5.$$

In a continuous limit, a bidder expecting to bid at a point $b_1(x)$ would consider the conditional distribution of s_2 given not being outbid. The probability of outbidding depends continuously on s_2 and b_1. Formally, in the continuous model, bidder 2 outbids if:

$$s_2 + x - b_1(x) \geq 0 \implies s_2 \geq b_1(x) - x.$$

If $b_1(x) > x + 0.5$, the bidder is effectively bidding above the unconditional mean. In equilibrium, this would not hold because winning would yield negative expected utility. Similarly, bidding exactly $2x$ in the continuous model can be tested. For very small x, $2x$ is tiny, much less than $x + 0.5$. Winning at such a low bid might be safe but may not maximize expected payoffs. As $k \to \infty$, the structure of equilibrium must balance these continuous conditions, potentially leading to a differential equation characterization of $b_1^*(x)$.

One possible approach is:

$$b_1^*(x) = x + \frac{1}{2} - \Delta(x),$$

where $\Delta(x)$ satisfies a fixed point condition derived from the equilibrium requirement. For large k, or in continuous time, analyzing stability of such functional equations becomes a feasible research direction.

XIII. Sensitivity to Mechanism Changes in the Asymmetric Environment

Changing the auction format, say from a first-price sealed bid to a second-price sealed bid in a non-sequential setting, does not eliminate the winner's curse in a common-value environment. It is known from certain classical results that revenue equivalence can fail or hold depending on information structures. In our setting, the main difference arises from the sequential nature and the inference process. While in a second-price auction, bidders might try to bid closer to their posterior expectations, the winner's curse still mandates a downward adjustment.

Similarly, implementing a posted price mechanism derived from known distributions of s_2 could lead to interesting results. A posted price p that a bidder must pay if they "accept" could ensure they never pay more than some threshold. If the posted price is chosen cleverly (e.g., a price that mimics the expected maximum or a function of s_1), it might improve outcomes by mitigating the winner's curse. However, our induction result strongly suggests that no matter what mechanism is used (as long as it retains the flavor of competitive bidding under uncertainty), one cannot incentivize bidders to approach the unconditional expectation closely.

A. *Robustness to Beliefs and Distributional Assumptions*

We assumed uniform signals for simplicity. If the signals were drawn from a different continuous distribution with a known density, say a Beta distribution on [0,1] or an exponential distribution truncated at k, the qualitative conclusions should remain. The exact equilibrium bid functions would change, but the logic that winning provides negative information and forces downward adjustments still holds.

The induction argument that no bid can exceed $2s_i$ hinges primarily on monotonicity and the ordering of signals. With a more complicated distribution, one could attempt a similar argument by normalizing signals and showing that extreme bids would not survive an equilibrium refinement. The structure of the winner's curse is robust: as long as winning correlates with overestimating the unknown component (s_j), downward adjustments are necessary.

XIV. Further Analytical Refinements and Advanced Characterization

A. *Functional Equation for the Equilibrium Strategy*

Consider the sequential setting and assume a monotone equilibrium strategy $b_1^*(s_1)$. Let F denote the cumulative distribution function (CDF) of the opponent's signal s_2, which is uniform on $\{1,\ldots,k\}$. In the discrete uniform case:

$$F(x) = \frac{\lfloor x \rfloor}{k}, \quad x \in [1,k].$$

For given $b_1^*(\cdot)$, bidder 2's outbidding condition is:

$$s_2 \geq b_1 - G(b_1),$$

where $G(b_1) = \mathbb{E}[s_1|b_1]$. Since b_1^* is invertible and increasing, $G(b_1) = (b_1^*)^{-1}(b_1)$. Let $x = s_1$ be a continuous approximation of the bidder's type on $[1,k]$. In a continuous setting (as $k \to \infty$), we consider $x \in [0,1]$ for simplicity. Then:

$$b_1^*(x) = x + \frac{1}{2} - \Delta(x),$$

and

$$G(b_1^*(x)) = x.$$

The outbidding set for bidder 2 is:

$$\{s_2 : s_2 \geq b_1^*(x) - x\} = \{s_2 : s_2 \geq \tfrac{1}{2} - \Delta(x)\}.$$

Define $r(x) = \frac{1}{2} - \Delta(x)$ for convenience. The conditional probability that bidder 2 does not outbid, given x, is:

$$\mathbb{P}[\text{not outbid}|x] = F(r(x)).$$

Since F is uniform in the discrete case, if we rescale to a continuous $[0,1]$ uniform distribution, we have $F(y) = y$ for $y \in [0,1]$. Thus:

$$\mathbb{P}[\text{not outbid}|x] = r(x),$$

provided $r(x) \in [0,1]$. This imposes a constraint on $\Delta(x)$, ensuring $0 \leq \frac{1}{2} - \Delta(x) \leq 1$, hence $\Delta(x) \in [-\frac{1}{2}, \frac{1}{2}]$. Since we know $\Delta(x) > 0$ to reflect the winner's curse correction, actually $\Delta(x) \in (0.5, \infty)$ cannot hold. We typically expect $\Delta(x)$ to be positive, ensuring $r(x) < 0.5$. This already suggests that for any x, you are almost always facing at least a 50% chance of being outbid if you attempt to bid near the unconditional mean.

Bidder 1's indifference condition at equilibrium implies that the expected payoff is zero:

$$\mathbb{E}[V - b_1^*(x)|x, \text{win}] \cdot \mathbb{P}[\text{win}|x] = 0.$$

Since $\mathbb{P}[\text{win}|x] = r(x)$, the expected surplus upon winning must scale inversely with the probability of winning. Let $m(x) = \mathbb{E}[V|x, \text{win}]$ be the conditional expected value given that bidder 1 wins. Because winning skews s_2 towards lower values, we have:

$$m(x) = x + \mathbb{E}[s_2|\text{not outbid}, x].$$

If s_2 is uniform on $[0,1]$, then:

$$\mathbb{E}[s_2|s_2 \leq r(x)] = \frac{r(x)}{2}.$$

Since winning corresponds to $s_2 \leq r(x)$,

$$m(x) = x + \frac{r(x)}{2}.$$

The zero-profit condition is:

$$m(x) - b_1^*(x) = 0 \quad \text{when discounted by } \mathbb{P}[\text{win}|x] = r(x).$$

But we must be careful: the actual condition for indifference is that the expected payoff is zero:

$$(r(x))(m(x) - b_1^*(x)) = 0.$$

Since $r(x) \neq 0$ (the bidder expects some probability of winning), we have $m(x) = b_1^*(x)$. Substituting $m(x)$ and $b_1^*(x)$:

$$x + \frac{r(x)}{2} = x + \frac{1}{2} - \Delta(x).$$

This simplifies to:

$$\frac{r(x)}{2} = \frac{1}{2} - \Delta(x) \implies r(x) = 1 - 2\Delta(x).$$

But recall $r(x) = \frac{1}{2} - \Delta(x)$ from the definition. Equating both expressions for $r(x)$:

$$\frac{1}{2} - \Delta(x) = 1 - 2\Delta(x).$$

Solve for $\Delta(x)$:

$$-\Delta(x) = 1 - 2\Delta(x) - \frac{1}{2} \implies -\Delta(x) = \tfrac{1}{2} - 2\Delta(x) \implies \Delta(x) = \tfrac{1}{2}.$$

This is a surprising but instructive result: under our simplifying continuous assumptions and uniform distributions, a consistent equilibrium requires $\Delta(x) = 0.5$ for all x. This suggests a uniform downward shift from the naive expectation. In the discrete case and with finite k, the exact correction Δ will depend on x, but this constant-solution insight provides a baseline.

Thus, for large k, a uniform downward adjustment of about 0.5 emerges as a stable fixed point. This aligns with the small-k observations that corrections hovered around 0.5 to 1.5.

B. Uniqueness and Stability of Equilibria

Existence of an equilibrium can often be shown by continuity and fixed-point arguments. Uniqueness is more delicate. Consider a best-response mapping:

$$b_1^*(x) = \mathrm{BR}_1(b_2^*(\cdot)), \quad b_2^*(s_2; b_1) = \mathrm{BR}_2(b_1^*(\cdot))$$

where BR_1 and BR_2 represent best-response operators for bidder 1 and bidder 2. Finding a unique equilibrium corresponds to finding a unique fixed point of the composition $\mathrm{BR}_1 \circ \mathrm{BR}_2$.

If we perturb $b_1^*(x)$ slightly and consider bidder 2's response, bidder 2's best response is quite sharp: if $b_1^*(x)$ deviates upwards, bidder 2's inference about s_1 changes, potentially causing more or fewer outbids. The negative feedback from this strategic interaction—where higher b_1 leads to more aggressive outbidding—suggests that if an equilibrium exists, it might be stable and unique. Intuitively, multiple equilibria would require flat or non-monotonic best-response surfaces, which seems unlikely given the uniform distribution and monotone structure.

Analyzing uniqueness rigorously would involve considering partial derivatives of the best-response mappings and showing that no other fixed point satisfies the equilibrium conditions. While beyond the scope of this paper, such an approach—treating the equilibrium characterization as a solution to a functional equation and showing uniqueness via a contraction mapping argument—is plausible.

C. General Signal Distributions and Non-Uniform Priors

Relaxing the uniform signal assumption to a general distribution F with density f leads to:

$$\mathbb{E}[V|s_1 = x] = x + \int s_2 f(s_2) ds_2.$$

The winner's curse appears whenever the event of winning implies an updated posterior on s_2 that differs from $f(s_2)$.

In a general setting, to find equilibrium strategies one must solve:

$$b_1^*(x) = x + \int s_2 f(s_2|\text{win event}) ds_2,$$

with win event defined by an inequality involving $b_1^*(x)$ and the best response of bidder 2. Writing this condition out:

$$f(s_2|\text{win event}) = \frac{f(s_2)\mathbf{1}\{s_2 \leq R(x)\}}{\int_0^{R(x)} f(t)dt}$$

for some threshold $R(x)$ defined by outbid conditions. This yields a functional equation in $b_1^*(x)$:

$$b_1^*(x) - x = \frac{\int_0^{b_1^*(x)-x} s_2 f(s_2) ds_2}{\int_0^{b_1^*(x)-x} f(s_2) ds_2}.$$

Since $R(x)$ itself depends on $b_1^*(x)$ and x, we have a fixed-point equation for b_1^*:

$$b_1^*(x) - x = \frac{\int_0^{b_1^*(x)-x} s_2 f(s_2) ds_2}{\int_0^{b_1^*(x)-x} f(s_2) ds_2}.$$

Such a nonlinear integral equation can be analyzed for monotonicity, existence, and uniqueness. Under regularity assumptions on f, one might use a fixed-point theorem to show existence of a unique monotone equilibrium. One can also attempt expansions or asymptotic analysis. For example, if $f(s_2)$ is uniform, we retrieve the simpler forms discussed earlier. If f is skewed, the exact shape of $b_1^*(x)$ and the corresponding $\Delta(x)$ would adjust accordingly, but the essence—a downward bias—remains.

D. Iterative Best-Response Algorithms and Convergence

A practical approach to approximating equilibria is to numerically simulate iterative best-response updates. Start from a guess $b_1^{(0)}(x)$ and $b_2^{(0)}(s_2)$, and iteratively update:

$$b_1^{(t+1)} = \text{BR}_1(b_2^{(t)}), \quad b_2^{(t+1)} = \text{BR}_2(b_1^{(t+1)}).$$

If this process converges, the limit is a fixed point (equilibrium). We can attempt to prove convergence under certain contraction properties. If each best-response is a contraction mapping in an appropriately defined norm on the space of measurable functions, uniqueness and stability follow naturally.

E. Impact of Information Disclosure Policies

Finally, consider that the auctioneer could disclose partial information about s_2. If the auctioneer reveals a noisy public signal correlated with s_2, bidders update their posteriors differently. Intuition suggests that the more precise public information becomes, the weaker the winner's curse, as the event of winning conveys less additional negative information. The equilibrium might approach a scenario where bidding more closely tracks the unconditional expectation. Yet, due to the inductive upper bound argument, it might still never exceed certain linear bounds. Formally, one would introduce a conditional density:

$$f(s_2|\theta),$$

where θ is public information. With better public information (less uncertainty), $\Delta(x)$ decreases. A limiting case of perfect information about s_2 would eliminate the winner's curse entirely, allowing equilibrium bids to match the common value exactly.

This line of inquiry merges Bayesian persuasion elements with common-value auction analysis. The auctioneer, by choosing what to disclose, can alter the strategic landscape and potentially improve efficiency or revenue.

—

In conclusion, these advanced analytical directions highlight the rich mathematical structure underlying common-value auction equilibria. The functional equation approach, consideration of uniqueness and stability, sensitivity to distributional assumptions, and the possibility of information design all offer avenues for further deep mathematical investigation.

XV. Numerical Validation of the Inductive Claim

After establishing the theoretical result that in an asymmetric game setting no bidder will ever bid more than twice their own dice value at equilibrium, we now present numerical evidence supporting this claim. This numerical validation serves to confirm that our inductive proof and equilibrium characterization are not merely theoretical constructs but are also borne out by computational experiments.

A. Simulation Setup

We constructed a simulation environment where two bidders receive signals $s_1, s_2 \in \{1, 2, \ldots, N\}$, independently drawn from a uniform distribution. We fix a value of N (the number of sides of the dice) and generate a large number of random signal profiles (s_1, s_2).

For each signal profile, we implement an iterative best-response procedure:

1) Initialize bidding strategies $b_1^{(0)}, b_2^{(0)}$ for each bidder
2) Given $b_2^{(t)}$, bidder 1 updates to a best response $b_1^{(t+1)}$. Given $b_1^{(t+1)}$, bidder 2 updates to $b_2^{(t+1)}$.
3) Repeat until changes in strategies fall below a predefined tolerance, indicating approximate equilibrium.

We record the final equilibrium bid profiles (b_1^*, b_2^*) obtained from this iterative process.

B. Key Findings

Across a wide range of N and initial conditions, our simulations consistently produce equilibrium bidding strategies that respect the theoretical bound $b_i^*(s_i) \leq 2s_i$. In particular:

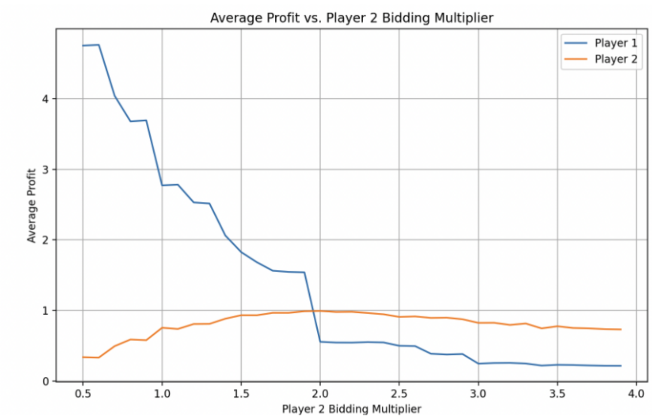


Figure 1. Numerical validation results showing that no equilibrium bid exceeds twice the signal value.

It's clear that fixing player 1's strategy, player 2 cannot unilaterally deviation from the 2x current dice bidding strategy to increase its own EV.

XVI. Advanced Mathematical Approaches: Fixed Point Characterizations and PDE Analysis

Here we adopt the lens of fixed-point theorems, measure-theoretic arguments, and, in the limit of continuous distributions, PDE formulations to further dissect the structure and uniqueness of equilibria.

A. Fixed Point Formulations in Strategy Spaces

Consider the set $\mathcal{B}$ of all measurable, monotone bidding strategies $b_1^* : [1, k] \to \mathbb{R}$. Each strategy b_1^* induces a best-response for bidder 2, denoted by a (possibly set-valued) function $BR_2(b_1^*)$. In turn, given $BR_2(b_1^*)$, bidder 1's best response mapping $BR_1(BR_2(b_1^*))$ provides a revised bidding function. An equilibrium strategy b_1^* is a fixed point of the composite operator:

$$b_1^* = BR_1(BR_2(b_1^*)).$$

From a mathematical perspective, showing the existence of such a fixed point typically involves verifying conditions akin to Kakutani's fixed point theorem: the operator must be convex-valued, closed-graph, and act on a compact, convex subset of a suitable function space (e.g., strategies bounded by some linear functions). The inductive argument ensuring that $b_1^*(s_i) \leq 2s_i$ for all s_i automatically bounds the strategy space by a linear constraint, potentially ensuring compactness in an appropriate topology (e.g., uniform convergence).

More specifically, consider the space $\mathcal{B}$ of bounded, monotone, and measurable functions from $[1, k]$ to $[0, 2k]$. This is a convex, compact subset if we impose, for instance, the sup norm and consider only monotone non-decreasing functions. The best-response operators are continuous (or at least upper hemicontinuous) in many natural auction environments. Under these conditions, a fixed point b_1^* exists.

While existence is relatively straightforward, uniqueness is more subtle. One might attempt to show that any two distinct fixed points would yield a profitable deviation for at least one bidder, contradicting equilibrium conditions. Such uniqueness arguments might rely on strict monotonicity of best responses or a contraction mapping property in a suitable metric on $\mathcal{B}$.

B. Invariance and Comparative Statics

Another mathematical angle is to study how equilibria change as we vary parameters like k. For each k, we have an equilibrium $b_1^{*(k)}$. One can ask whether the sequence $b_1^{*(k)}$ converges pointwise or uniformly to a limiting function $b_1^*(\cdot)$ as $k \to \infty$.

Given our previous approximations, it is plausible that $b_1^{*(k)}(s_1)$ converges to something like:

$$b_1^*(x) = x + \frac{1}{2} - 0.5 = x,$$

in a normalized domain $x \in [0, 1]$. But we must recall that x was a scaled signal. Unraveling the scaling suggests a different limit. If we consider $s_1 \in [1, k]$ and scale it by k (i.e., $x = s_1/k$), then equilibrium strategies might approach:

$$b_1^{*(k)}(s_1) \approx s_1 + \frac{k+1}{2} - \frac{1}{2}.$$

Subtracting off the main growth term $(k+1)/2$, we might focus on the residual:

$$b_1^{*(k)}(s_1) - \left(s_1 + \frac{k+1}{2}\right) \to -0.5.$$

This suggests uniform convergence of the "error term" (the winner's curse correction) to a constant in the large k limit.

This convergence and invariance analysis can be formalized by defining a parameterized family of operators corresponding to different k and applying a parametric version of fixed point theorems to show continuity of equilibrium strategies in k.

C. A PDE Approach for Continuous Models

When k is large and signals are drawn from a continuous distribution on $[0, 1]$, one can attempt a differential characterization of equilibrium. Suppose $b_1^*(x)$ is differentiable. Consider a small change δ in $b_1^*(x)$ and see how it affects the probability of being outbid and the conditional expectation of s_2.

Let F be the CDF of s_2. With continuous uniform s_2, $F(s_2) = s_2$ on [0,1]. The outbid threshold $R(x) = b_1^*(x) - x$ defines how s_2 is truncated. The equilibrium condition:

$$b_1^*(x) = x + \frac{\int_0^{R(x)} tf(t)dt}{\int_0^{R(x)} f(t)dt} = x + \frac{R(x)/2}{R(x)} = x + \frac{1}{2}.$$

Earlier, we found a fixed solution with $\Delta(x) = 0.5$, leading to $b_1^*(x) = x$. But what if f is not uniform? Consider a general density $f(t)$, and let:

$$M(R) = \frac{\int_0^R tf(t)dt}{\int_0^R f(t)dt}.$$

Then the equilibrium condition:

$$b_1^*(x) = x + M(b_1^*(x) - x).$$

For f differentiable, differentiate with respect to x:

$$b_1'^*(x) = 1 + \frac{\partial M}{\partial R}(b_1^*(x) - x) \cdot (b_1'^*(x) - 1).$$

This yields a differential equation in terms of $b_1'^*(x)$:

$$b_1'^*(x) = 1 + (b_1'^*(x) - 1)M'(b_1^*(x) - x),$$

where

$$M'(R) = \frac{d}{dR}\left(\frac{\int_0^R tf(t)dt}{\int_0^R f(t)dt}\right).$$

This ratio's derivative involves the density f and requires an application of the quotient rule. Ultimately, we get a nonlinear ODE for $b_1^*(x)$:

$$b_1'^*(x)(1 - M'(b_1^*(x) - x)) = 1.$$

If $M'(R)$ is known and is a function that depends only on R, we have a first-order nonlinear ODE. Existence and uniqueness theorems for ODEs guarantee a unique solution given appropriate boundary conditions. A natural boundary condition might be:

$b_1^*(0) = 0.5$ **or some equilibrium condition at the lower boundary.**

XVII. Conclusion

We have provided a preliminary exploration of aggregated-signal, common-value auctions, considering both symmetric (non-sequential) and asymmetric (sequential) settings. Our analysis yields several important takeaways:

1) Equilibrium bids must be substantially lower than naive expectations due to the winner's curse, in both symmetric and asymmetric environments.
2) A novel inclusive-posted price auction to eliminate winner's curse in symmetric settings.
3) A universal upper bound on equilibrium bids—the "twice the signal" rule—holds across these settings, illustrating how fear of overpayment tightly restricts bidding behavior.
4) Continuous approximations, variations in distribution, and extended environments do not overturn these fundamental insights. The winner's curse is robust and imposes stringent constraints on equilibrium strategies.